

Network Layer: Control Plane (Structured Notes)

1. Control Plane Overview

- Responsible for routing decisions and network configuration.
- Determines paths between source and destination.
- Separated from data plane, which performs packet forwarding.

2. Control Plane Architectures

Traditional (Per-router):

- Each router runs its own routing algorithm.
- Fully distributed approach.

Software Defined Networking (SDN):

- Centralized controller computes routing.
- Routers execute forwarding decisions.
- Separation between control and data plane.

3. Routing vs Forwarding

- Forwarding: local decision using forwarding table.
- Routing: global path computation across network.
- Routing algorithms populate forwarding tables.

4. IP Routing Basics

- Direct forwarding: same network.
- Indirect forwarding: via router (next hop).
- Netmask determines routing decision.

5. Longest Prefix Match

- Used when multiple routing entries match.
- Selects route with most specific prefix.
- Key principle of IP routing.

6. Classless Addressing (CIDR)

- Replaces class-based addressing.
- Uses flexible prefix lengths.
- Requires longest prefix match for routing.

7. Routing Algorithms

- Link State (LS): global knowledge of network.
- Distance Vector (DV): local knowledge with neighbors.

8. Link State Algorithm (Dijkstra)

- Each router knows full topology.

- Computes shortest paths using Dijkstra.
- Fast convergence but higher overhead.

9. Distance Vector Algorithm (Bellman-Ford)

- Routers exchange information with neighbors.
- Iterative updates based on Bellman-Ford equation.
- Issues: routing loops, count-to-infinity, slow convergence.

10. LS vs DV Comparison

- LS: faster convergence, higher complexity.
- DV: simpler but slower and less stable.
- Different trade-offs in scalability and robustness.

11. Autonomous Systems (AS)

- Internet divided into autonomous systems.
- Intra-AS routing within a domain.
- Inter-AS routing between domains.

12. Intra-AS Protocols

RIP:

- Distance vector protocol.
- Uses hop count metric.
- Limited scalability and slow convergence.

OSPF:

- Link state protocol.
- Uses Dijkstra algorithm.
- Hierarchical and scalable.

13. Inter-AS Routing: BGP

- Path vector protocol.
- Uses TCP for communication.
- Exchanges routing information between AS.
- Supports policy-based routing.

14. Key Concepts Summary

- Control plane manages routing logic.
- Forwarding is local, routing is global.
- Longest prefix match is fundamental.
- OSPF used inside AS, BGP between AS.